

Research Paper Implementation

Strategy: Explainable AI for Alzheimer's Diagnosis

The diagnostic trajectory for Alzheimer's Disease (AD) is transitioning from subjective neuropsychological assessments to computational biomarkers. While deep learning models like 3D-CNNs show high accuracy, their "black-box" nature limits clinical trust. This document outlines a comprehensive research plan to implement a diagnostic assistant using Squeeze-and-Excitation (SE) networks, Grad-CAM, and SHAP.

Part I: Recent Literature Survey (2023–2025)

The following papers focus on high-performance classification using attention mechanisms and 3D architectures, excluding the specific XAI layers, to provide a performance benchmark.

1. IEEE Publications

- **Ramineni et al. (2025):** "Attention-Enhanced Dual-Path CNN for Early Alzheimer's Detection." This study utilizes a dual-path architecture combining a domain-specific SNeurodCNN branch with an Inception-v4 backbone enhanced by Convolutional Block Attention Modules (CBAM) and SE blocks. It achieved 98.9% accuracy on the ADNI dataset by optimizing parasagittal slice orientations.
- **Zhou et al. (2025):** "Multi-Modal Diagnosis of Alzheimer's Disease Using Interpretable Graph Convolutional Networks." Published in *IEEE Transactions on Medical Imaging*, this work proposes a sparse GCN framework to detect salient Regions of Interest (ROIs) and connectivity abnormalities.¹
- **OmniBrain Framework (2025):** A multimodal model that fuses structural MRI, radiomics, and cognitive scores. It introduces a "Modality-Aware Attention Masking" strategy to handle missing data during inference, ensuring robustness in real-world clinical scenarios.

2. Springer Nature Publications

- **Alsowail & Al-Shehari (2025):** "Multi-stage Alzheimer's Disease Classification Using Squeeze Excitation Light Channel Network." This paper introduces a "Random Lyrebird" algorithm combined with SE blocks to achieve high sensitivity and precision in early detection.
- **Odusami et al. (2025):** "A Squeeze and Excitation Framework Utilizing ResNet-152 for Alzheimer's Disease Dementia Classification." This study demonstrates that incorporating SE blocks into deep residual networks recalibrates channel-wise features, reaching an overall accuracy of 99.00%.²
- **Hu et al. (2025):** "Alzheimer's Disease Diagnosis by 3D-SEConvNeXt." Published in the

Journal of Big Data, this research transforms the ConvNeXt model into a 3D architecture and adds 3D-SE attention to improve early-stage classification.

3. Elsevier & Other High-Impact Sources

- **Zia-Ur-Rehman et al. (2025):** "Advanced Convolutional Neural Network with Attention Mechanism for Alzheimer's Disease Classification." Published in *Computers in Biology and Medicine*, this work uses a connection-wise attention model and a dedicated MRI Segmentation Block (MSB) for automated skull stripping.
- **Heliyon (2024):** "Improved Neural Network with Multi-Task Learning for Alzheimer's Disease Classification." This study proposes ADNet, which replaces standard convolutions with depthwise separable convolutions and integrates SE modules to enhance feature extraction efficiency.
- **PLOS ONE (2025):** "Alzheimer's Disease Image Classification Based on Enhanced Residual Attention Network (ERAN)." This research combines residual learning with soft thresholding and SE modules to filter noise and focus on lesion areas, achieving 99.36% accuracy.³

Part II: Detailed Implementation Plan

1. Datasets (Open-Access & Free)

To ensure publishability, the research must utilize standardized, longitudinal datasets.

- **ADNI (Alzheimer's Disease Neuroimaging Initiative):** The gold standard for multimodal research. It provides 3D T1-weighted MRI, cognitive scores (MMSE, CDR), and genetic data (APOE- ϵ 4).
- **OASIS-3:** Latest version of the Open Access Series of Imaging Studies. It contains over 2,000 MR sessions and is ideal for training deep 3D-CNNs due to its large sample size.
- **MIRIAD:** A smaller, specialized dataset for tracking rapid atrophy in symptomatic AD patients, useful for validation.⁴

2. Software Tools & Frameworks

- **MONAI (Medical Open Network for AI):** Built on PyTorch, MONAI provides specialized 3D loaders, medical-grade transforms (intensity scaling, cropping), and optimized 3D-CNN implementations.⁵
- **ANTsPy / FSL:** Used for high-precision preprocessing, including N4 bias field correction, skull stripping, and MNI152 registration.⁷
- **Captum:** A unified library for PyTorch that will be used to implement Grad-CAM for the imaging branch.⁹
- **SHAP Python Library:** Used to calculate Shapley values for the tabular clinical data branch.¹⁰

3. Step-by-Step Workflow

1. **Preprocessing (The Foundation):** Use ANTsPy to perform N4 intensity correction and register scans to the **MNI152 1mm template**. Resize volumes to $128 \times 128 \times 128$ and normalize voxel intensities to $[-1, 1]$.
2. **Feature Extraction (Imaging):** Build a **3D-SE-ResNet-18** or **3D-SE-DenseNet-121**. The SE blocks should be placed after each residual or dense block to recalibrate channel weights based on the global spatial context.
3. **Feature Extraction (Clinical):** Process tabular features (Age, MMSE, CDR, APOE- ϵ 4) through an **XGBoost** model. These models handle small, structured datasets more effectively than neural networks.
4. **Multimodal Fusion:** Concatenate the global average pooled features from the 3D-CNN with the clinical model's output. Pass this through a final MLP layer for 4-way classification (CN, EMCI, LMCI, AD).

4. Strategies to Increase Accuracy

- **MedicalNet Pre-training:** Instead of ImageNet, initialize the 3D-CNN with weights from **MedicalNet**, which are pre-trained on massive 3D medical datasets (e.g., LiTS, BraTS).⁹
- **Focal Loss:** Address class imbalance (where healthy subjects often outnumber AD subjects) by using Focal Loss, which focuses the network on "hard" examples (e.g., MCI cases transitioning to AD).
- **3D Data Augmentation:** Use MONAI's RandFlipd, RandRotate90d, and RandZoomd to increase spatial variability and prevent overfitting.

Part III: Explainable AI Integration

1. Grad-CAM for Visual Interpretability

Grad-CAM will be applied to the final convolutional layer of the 3D-CNN. It generates 3D heatmaps that highlight neuroanatomical regions like the **hippocampus** and **entorhinal cortex**.

- **Validation:** Map the Grad-CAM heatmaps to the **Automated Anatomical Labeling (AAL3) atlas**. This allows the AI to output a quantitative report (e.g., "Prediction based on 80% activation in the Right Hippocampus").

2. SHAP for Clinical Interpretability

SHAP is used to explain the tabular branch. It assigns an importance value to each clinical marker (e.g., "MMSE score contributed -0.4 to the AD prediction").¹²

- **Holistic Explanation:** By combining Grad-CAM (spatial) and SHAP (tabular), the system

justifies its diagnosis through both visual evidence of atrophy and symbolic clinical reasoning, creating a true "Diagnostic Assistant."

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